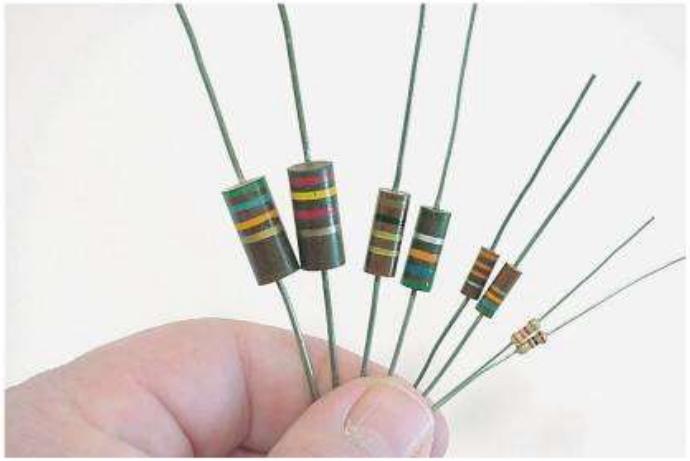
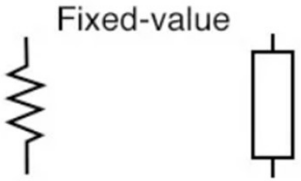
Resistor Circuits

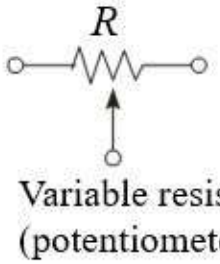
- Resistors
- Series Circuit or Voltage Divider Circuit
- Parallel Circuit or Current Divider Circuit
- Power Dissipation in Resistor Circuit

Resistors

- Components designed to have a specific amount of resistance are called *resistors*.
- Resistors limit current in a circuit
- Resistors are of two types – **fixed-value** resistors, **variable** resistors (e.g., potentiometer)

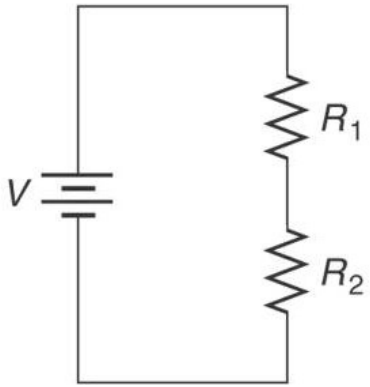








Series Resistor Circuits



- In series resistor circuit, current is same, voltage divides.

$$V_S = V_1 + V_2 + \ldots + V_n$$

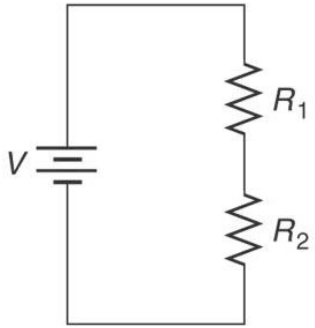
- Total Series Resistance is sum of individual resistances

$$R_T = R_1 + R_2 + \ldots + R_n$$

Where, R_T = the total circuit resistance

Voltage Divider Circuit

- The Voltage Divider Relationship
  - Voltage Divider – often used to describe a series circuit

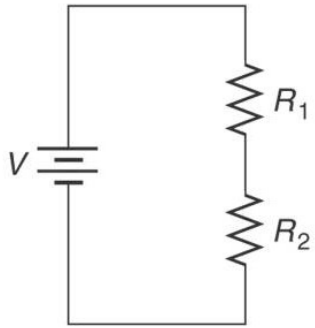


Voltage Divider Circuit

Voltage Divider Rule:

- The voltage drop across any given resistor in a series circuit is equal to the ratio of that resistor to the total resistance, multiplied by source voltage.

$$\frac{V_n}{V_S} = \frac{R_n}{R_T} \Rightarrow V_n = V_S \frac{R_n}{R_T}$$



$$V_1 = \left(\frac{R_1}{R_T}\right) V_S$$

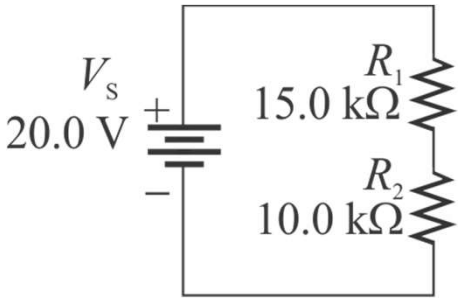
$$V_2 = \left(\frac{R_2}{R_T}\right) V_S$$

where

R_n = the resistor of interest

V_n = the voltage drop across R_n
(where n is the component number)

Q. What is the voltage across R_2?



The total resistance is 25 kΩ.
Applying the voltage divider formula:

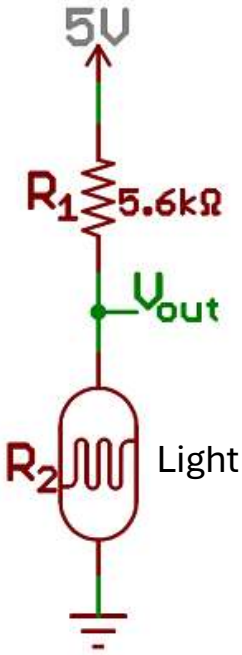
$$V_2 = \left(\frac{R_2}{R_T}\right) V_S = \left(\frac{10\ \text{k}\Omega}{25\ \text{k}\Omega}\right) 20\ \text{V} = \quad 8.0\ \text{V}$$

What is the voltage across R_1?

$$V_1 = Vs - V_2$$

Notice that 40% of the source voltage is across R2, which represents 40% of the total resistance.

Voltage Divider Circuit Application



$$V_{out} = \frac{R_2}{R_1 + R_2} \times V_{in}$$

Light Sensor (LDR)

- Two resistors in series: fixed R1 and variable R2 (LDR).
- The output voltage

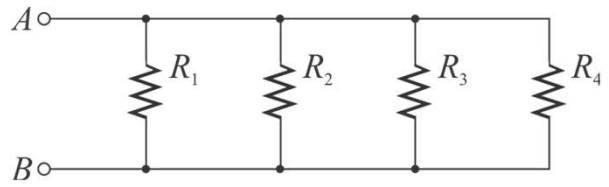
Light Level	R_2 (Sensor)	R_1 (Fixed)	Ratio $R_2/(R_1+R_2)$	V_{out}
Light	1kΩ	5.6kΩ	0.15	0.76 V
Dim	7kΩ	5.6kΩ	0.56	2.78 V
Dark	10kΩ	5.6kΩ	0.67	3.21 V

LDR (Light Dependent Resistor) Behavior:

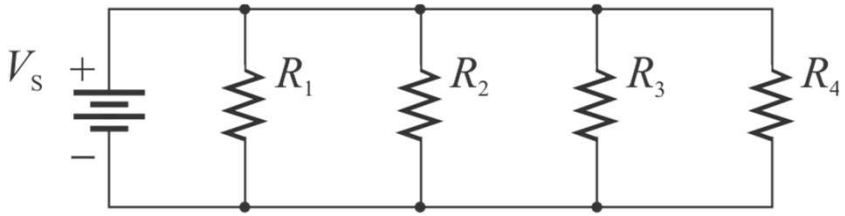
Bright light → $R2$ low → $Vout$ small.
Darkness → $R2$ high → $Vout$ larger.

Resistors in Parallel

- Resistors that are connected between the same two points are said to be in **parallel**.



- A ***parallel circuit*** has more than one current path (branch) connected to a common voltage source.

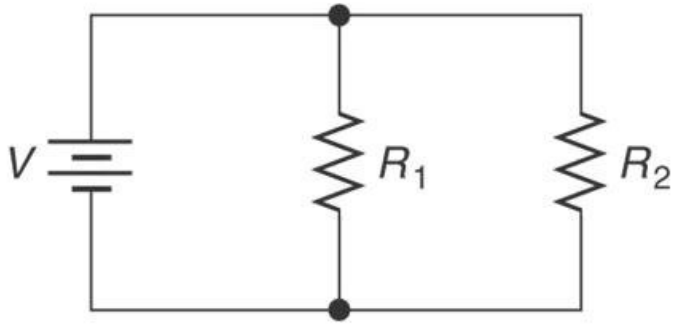


- Current through each branch is determined by the source voltage and the resistance of the branch.

$$V_S = V_1 = V_2 = \ldots = V_n$$

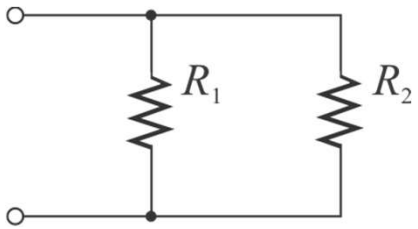
$$I_T = I_1 + I_2 \ldots + I_n$$

Total Resistance in Parallel Resistor Circuits



$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}}$$

Two Parallel Resistors



The special case of resistance of two parallel resistors can be found by:

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

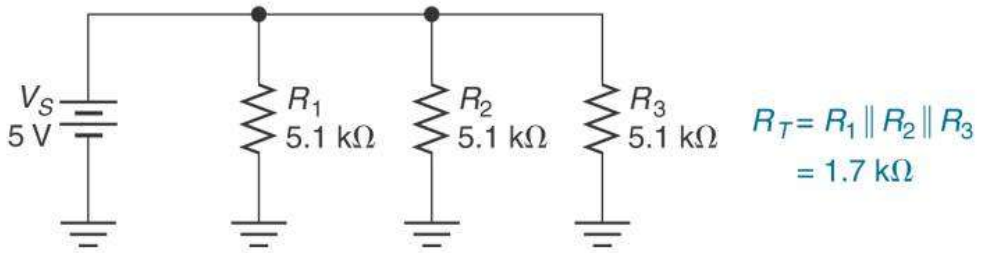
which can be reduced to

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

This is called the product-over-sum rule.

Calculate the total resistance if R_1 = 27 kΩ and R_2 = 56 kΩ 18.2 kΩ

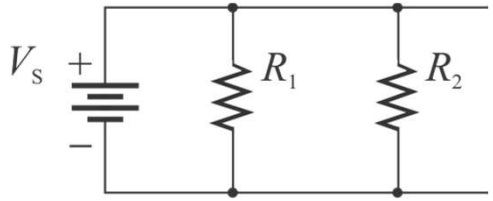
- Calculating Total Resistance When Branch Resistances are Equal – Divide the value of one resistor by the number of branches



$$R_T = \frac{5.1\text{k}\Omega}{3} = 1.7\text{k}\Omega$$

Parallel Resistor Circuit or **Current Divider Circuit**

- A Current Divider is a parallel circuit in which the **source current divides** among a number of parallel connected paths.
- When current enters a node (junction) it divides **inversely proportional to each branch resistance.**



$$I_k = I_T \cdot \frac{\frac{1}{R_k}}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}}$$

$$I_k = I_T \cdot \frac{R_{total}}{R_k}$$

$I_T = I_{R1} + I_{R2}$

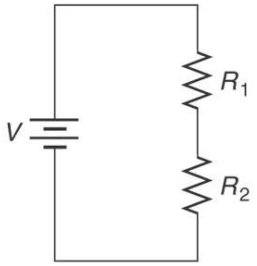
$V_{R1} = V_{R2} = V_{R3}$

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

Series Vs Parallel Circuits

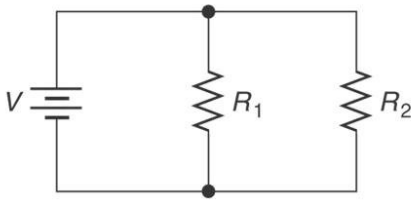
Series:

- **Single current path**: All components share the same current.
- **Voltage division**: The total voltage of the source is divided among the components based on their resistance.
- **Current continuity**: The current is the same through every component, indicating that if one component fails, the entire circuit stops functioning.

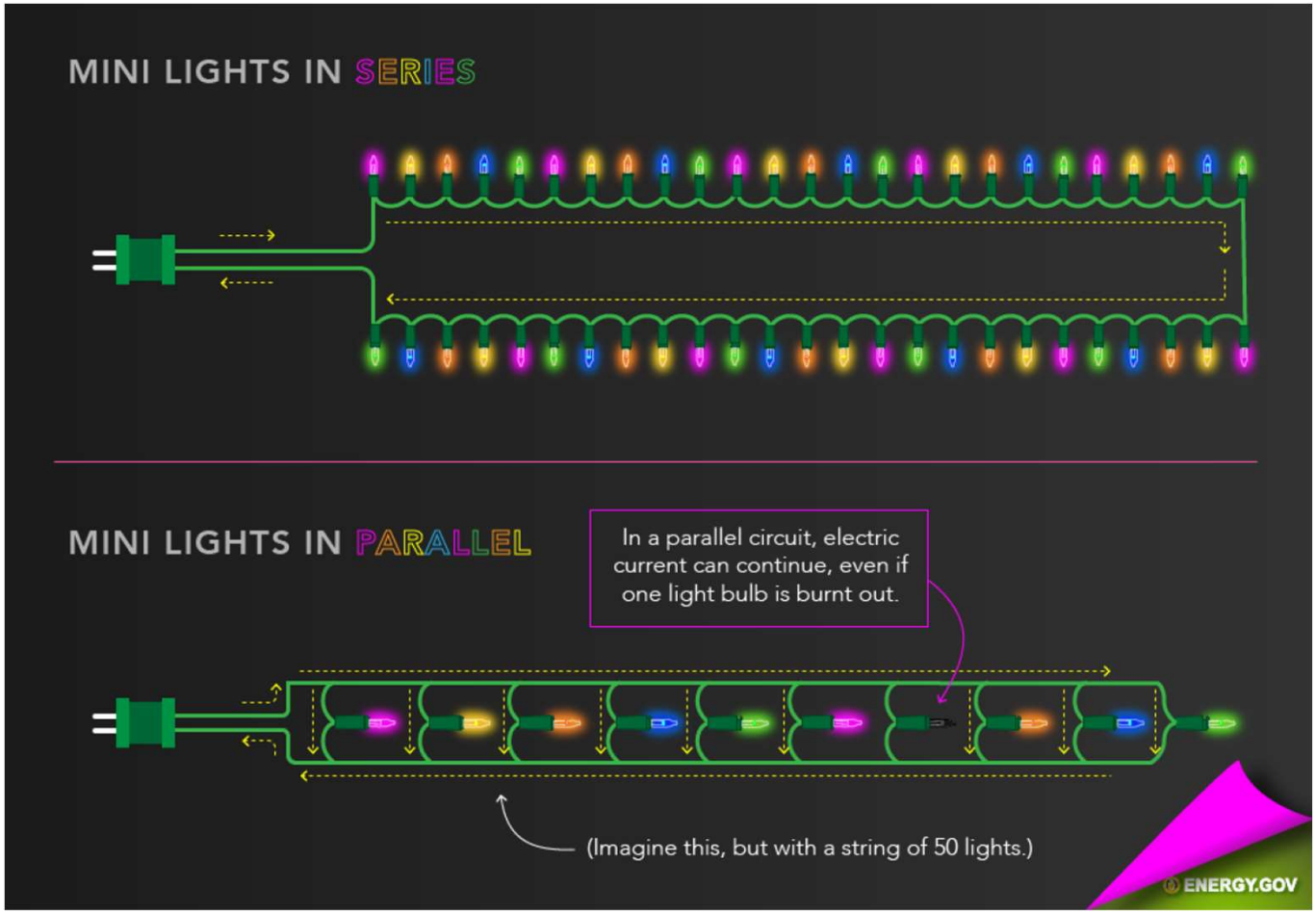


Parallel

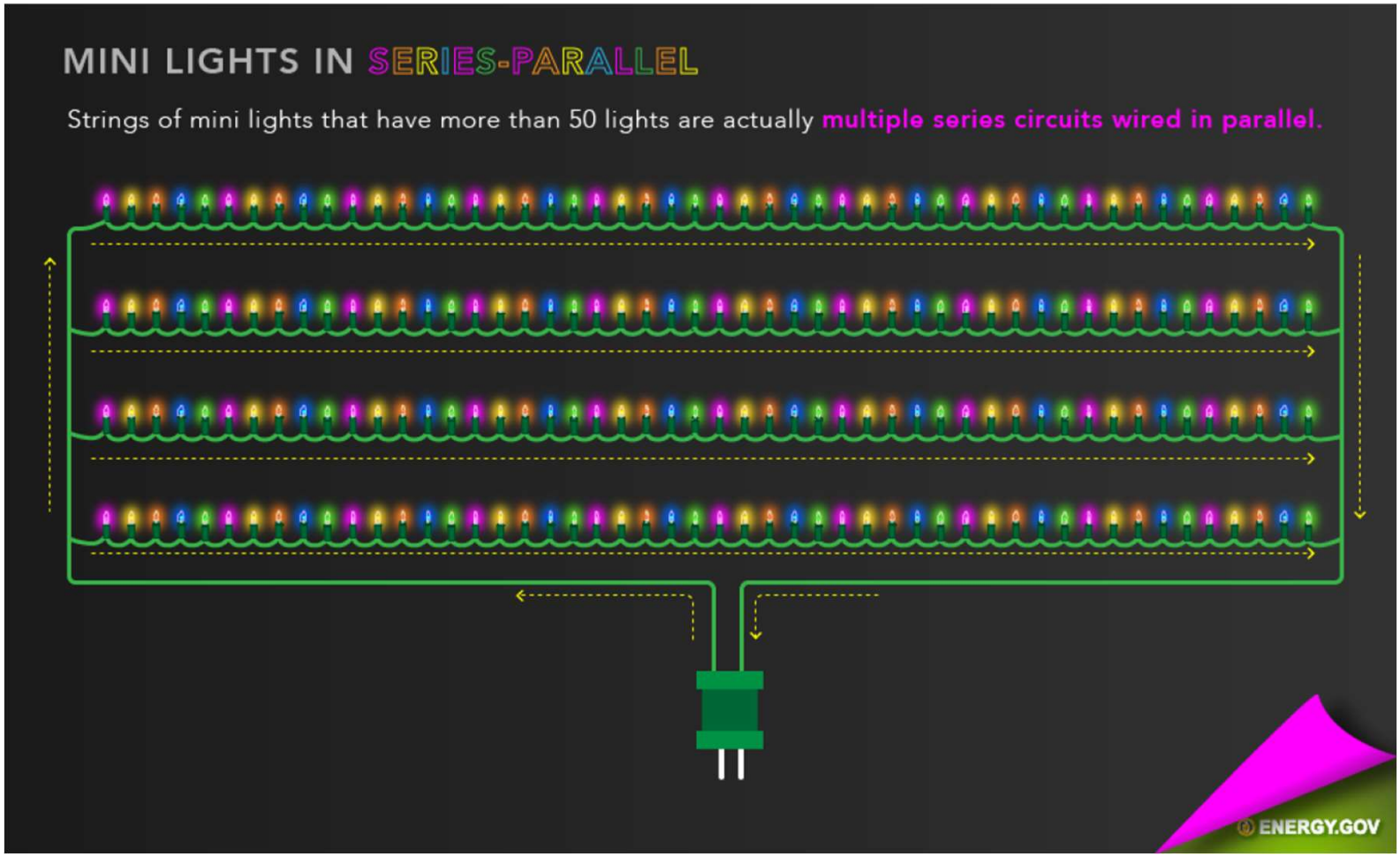
- **Multiple current paths:** Current divides among the various branches depends on the resistance of each branch.
- **Uniform voltage:** All components share the same voltage, which is equal to the voltage of the supply.
- **Independent operation:** The failure of one component does not stop the flow of current in other paths



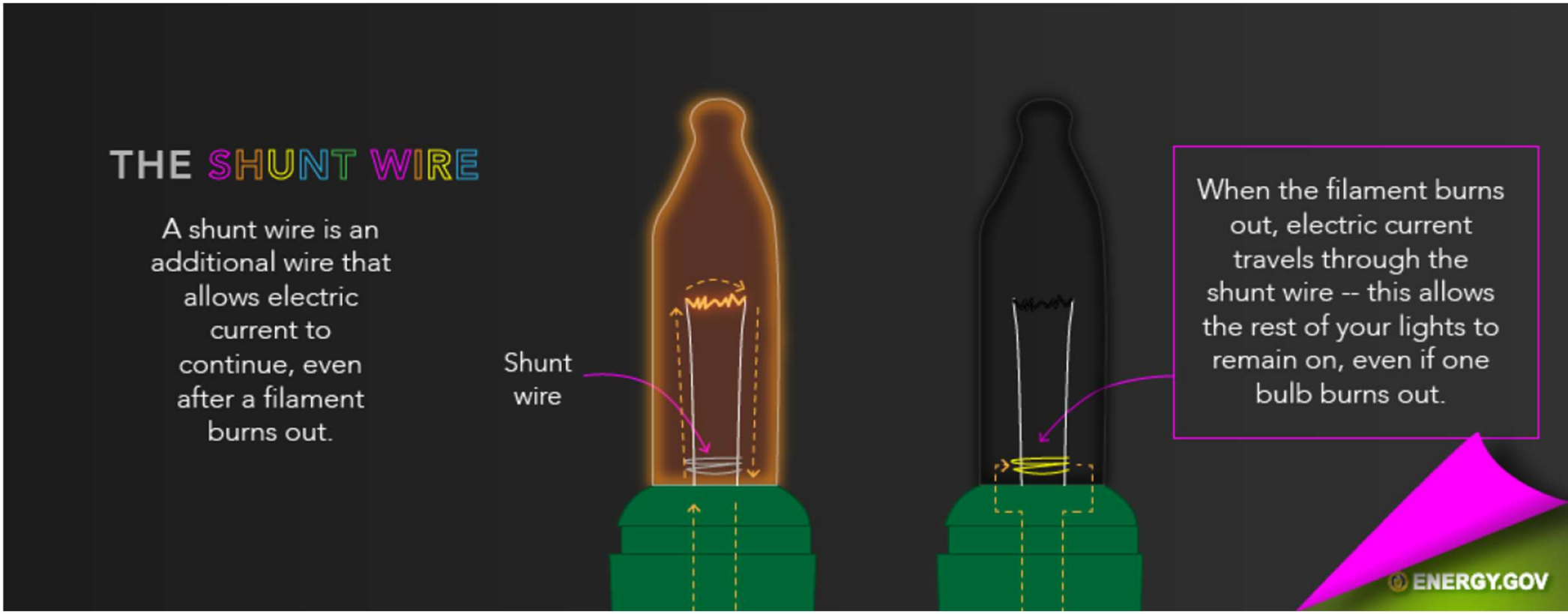
Holiday lights circuits



PC: https://www.energy.gov/



PC: https://www.energy.gov/



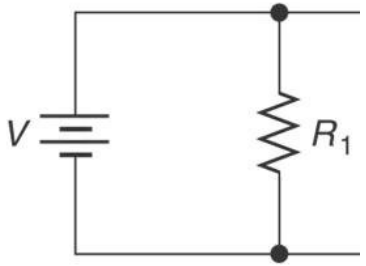
PC: https://www.energy.gov/

Power in Electric Circuit

- Power is the **amount of energy** used **per unit time.**
- Unit of Power is **watt (W)**

Power Calculation: $P = VI$

- What is the power dissipation in the resistor?



Power dissipation in an electric circuit is seen as heat given off by the resistance. Power dissipation is equal to the power produced by the voltage source.

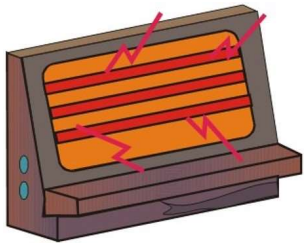
Power in an Electric Circuit

In electrical circuit, the rate energy is dissipated can be determined from any of three forms of the power formula.

$$P = I^2R \qquad P = VI \qquad P = \frac{V^2}{R}$$

Together, the three forms are called **Watt's law.**

Q. What power is dissipated by a heater that draws 12 A of current from a 120 V supply?



$$P = IV$$
$$= (12\ \text{A})(120\ \text{V})$$
$$= 1440\ \text{W}$$